Lean Canvas

Autonomous Learning Mentor — Autonomous Multi-Agent AI Mentor for SDG 4: Quality Education

Problem → Solution → Value → Advantage → Customers

EXISTING ALTERNATIVES

- Generic MOOCs (Coursera, Udemy) with static curricula
- 1:1 human tutoring — effective but expensive and hard to scale
- Flashcard / spaced-repetition apps with no real teaching layer

1. PROBLEM

- Fixed, one-size-fits-all courses that never adjust to pace, skill level, or real weak spots
- Quiz results rarely reshape what gets taught next — no real feedback loop
- Tutoring, testing, and scheduling live in separate, disconnected tools

2. SOLUTION

- Nine specialized AI agents (Goal, Planner, Tutor, Quiz, Progress, Adaptive, Revision, Reminder, Memory) run through one LangGraph workflow
- Roadmap re-plans automatically from real quiz performance

KEY METRICS

- Roadmap completion rate
- Average quiz score improvement over time
- Weekly learning streak / active days
- Weak-topic → strong-topic conversion rate

3. UNIQUE VALUE PROPOSITION

An autonomous AI mentor — not a chatbot — that plans, teaches, tests, and adapts a learner's entire journey automatically.

HIGH-LEVEL CONCEPT

"Like a personal tutor who never sleeps and never forgets your progress."

4. UNFAIR ADVANTAGE

- Real multi-agent LangGraph architecture with conditional routing and checkpointing — not a single prompt
- One shared memory across planning, tutoring, testing, and scheduling

CHANNELS

- Direct web app (Streamlit)
- Educational institution and bootcamp partnerships
- Open-source / GitHub community adoption
- SDG-aligned education programs and hackathons

EARLY ADOPTERS

- Hackathon participants and student developers
- Online learners already using AI study tools
- Coding bootcamp students preparing for interviews

5. CUSTOMER SEGMENTS

- Self-directed learners targeting technical roles (coding, data structures, ML)
- Students needing personalized supplementary tutoring
- Bootcamps and educators wanting an adaptive-learning layer

COST STRUCTURE

- LLM inference costs (Groq API usage across all nine agents)
- Cloud hosting and compute for the app, database, and vector store
- Ongoing development and agent-prompt maintenance

REVENUE STREAMS

- Freemium: free core roadmap, paid premium tutoring depth and advanced analytics
- Institutional / classroom licensing for bootcamps and schools
- Individual subscription for unlimited AI tutoring and quizzes